

SUPPORT2 Dataset: Data Cleaning & Preparation

AAI-500 Applied Statistics for AI | Final Team Project

Notebook Objectives

This notebook covers the **Data Cleaning & Preparation** phase of the project pipeline:

1. Initial data inspection and profiling
 2. Column renames per PEP-8
 3. Engineering the binary target variable (`death_180d`)
 4. Data type casting and ordinal/nominal encoding
 5. Missing data analysis and imputation
 6. Dropping unused columns
 7. Documenting target / feature / outcome
 8. Export of the cleaned dataset for EDA and modeling
-

Section 0: Setup & Imports

Output path : C:\Users\mw...Documents\USD-AAI\ai-500\final\usd-ai-500-final\data\support2_cleaned.csv

Section 1: Load Data & Initial Inspection

Shape: 9,105 rows x 48 columns

	id	age	death	sex	hospdead	slos	d.time	dzgroup	dzclass	nur
0	1	62.84998	0	male	0	5	2029	Lung Cancer	Cancer	
1	2	60.33899	1	female	1	4	4	Cirrhosis	COPD/CHF/Cirrhosis	
2	3	52.74698	1	female	0	17	47	Cirrhosis	COPD/CHF/Cirrhosis	
3	4	42.38498	1	female	0	3	133	Lung Cancer	Cancer	
4	5	79.88495	0	female	0	16	2029	ARF/MOSF w/Sepsis	ARF/MOSF	

5 rows × 48 columns

<class 'pandas.DataFrame'>

RangeIndex: 9105 entries, 0 to 9104

Data columns (total 48 columns):

#	Column	Non-Null Count	Dtype
0	id	9105 non-null	int64
1	age	9105 non-null	float64
2	death	9105 non-null	int64
3	sex	9105 non-null	str
4	hospdead	9105 non-null	int64
5	slos	9105 non-null	int64
6	d.time	9105 non-null	int64
7	dzgroup	9105 non-null	str
8	dzclass	9105 non-null	str
9	num.co	9105 non-null	int64
10	edu	7471 non-null	float64
11	income	6123 non-null	str
12	scoma	9104 non-null	float64
13	charges	8933 non-null	float64
14	totcst	8217 non-null	float64
15	totmcst	5630 non-null	float64
16	avtisst	9023 non-null	float64
17	race	9063 non-null	str
18	sps	9104 non-null	float64
19	aps	9104 non-null	float64
20	surv2m	9104 non-null	float64
21	surv6m	9104 non-null	float64
22	hday	9105 non-null	int64
23	diabetes	9105 non-null	int64
24	dementia	9105 non-null	int64
25	ca	9105 non-null	str
26	prg2m	7456 non-null	float64
27	prg6m	7472 non-null	float64
28	dnr	9075 non-null	str
29	dnrday	9075 non-null	float64
30	meanbp	9104 non-null	float64
31	wblc	8893 non-null	float64
32	hrt	9104 non-null	float64
33	resp	9104 non-null	float64
34	temp	9104 non-null	float64
35	pafi	6780 non-null	float64
36	alb	5733 non-null	float64
37	bili	6504 non-null	float64
38	crea	9038 non-null	float64
39	sod	9104 non-null	float64
40	ph	6821 non-null	float64
41	glucose	4605 non-null	float64
42	bun	4753 non-null	float64
43	urine	4243 non-null	float64
44	adlp	3464 non-null	float64
45	adls	6238 non-null	float64
46	sfdm2	7705 non-null	str
47	adlsc	9105 non-null	float64

dtypes: float64(31), int64(9), str(8)

memory usage: 3.3 MB

```
=====Binary Columns=====
['death', 'hospdead', 'diabetes', 'dementia']
```

1.1 Descriptive Statistics

	Count	Mean	Std Dev	Min	Q1 (25%)	Median (50%)
id	9105.0	4553.000000	2628.531434	1.000000	2277.000000	4553.000000
age	9105.0	62.650823	15.593710	18.041990	52.797000	64.856990
death	9105.0	0.681054	0.466094	0.000000	0.000000	1.000000
hospdead	9105.0	0.259198	0.438219	0.000000	0.000000	0.000000
slos	9105.0	17.863042	22.006440	3.000000	6.000000	11.000000
d.time	9105.0	478.449863	560.383272	3.000000	26.000000	233.000000
num.co	9105.0	1.868644	1.344409	0.000000	1.000000	2.000000
edu	7471.0	11.747691	3.447743	0.000000	10.000000	12.000000
scoma	9104.0	12.058546	24.636694	0.000000	0.000000	0.000000
charges	8933.0	59995.787811	102648.778198	1169.000000	9740.000000	25024.000000
totcst	8217.0	30825.867768	45780.820986	0.000000	5929.566400	14452.734400
totmcst	5630.0	28828.877838	43604.261932	-102.719970	5177.404300	13223.500000
avtisst	9023.0	22.610928	13.233248	1.000000	12.000000	19.500000
sps	9104.0	25.525872	9.899377	0.199982	19.000000	23.898438
aps	9104.0	37.597979	19.903852	0.000000	23.000000	34.000000
surv2m	9104.0	0.635870	0.248175	0.000000	0.507690	0.715942
surv6m	9104.0	0.520096	0.253343	0.000000	0.342957	0.574951
hday	9105.0	4.400000	9.126811	1.000000	1.000000	1.000000
diabetes	9105.0	0.195277	0.396436	0.000000	0.000000	0.000000
dementia	9105.0	0.032510	0.177359	0.000000	0.000000	0.000000
prg2m	7456.0	0.618769	0.298367	0.000000	0.500000	0.700000
prg6m	7472.0	0.499679	0.304367	0.000000	0.200000	0.500000
dnrday	9075.0	14.582590	19.944216	-88.000000	4.000000	9.000000
meanbp	9104.0	84.546408	27.687692	0.000000	63.000000	77.000000
wblc	8893.0	12.347677	9.266329	0.000000	7.000000	10.599609
hrt	9104.0	97.156711	31.559292	0.000000	72.000000	100.000000
resp	9104.0	23.330294	9.573801	0.000000	18.000000	24.000000
temp	9104.0	37.103341	1.251796	31.699220	36.195310	36.695310
pafi	6780.0	239.529070	109.665593	12.000000	155.093750	224.000000
alb	5733.0	2.950520	0.875944	0.399963	2.399902	2.899902
bili	6504.0	2.554463	5.318448	0.099991	0.500000	0.899902
crea	9038.0	1.770961	1.686041	0.099991	0.899902	1.199951

	Count	Mean	Std Dev	Min	Q1 (25%)	Median (50%)
sod	9104.0	137.568541	6.029326	110.000000	134.000000	137.000000
ph	6821.0	7.415364	0.080563	6.829102	7.379883	7.419922
glucose	4605.0	159.873398	88.391541	0.000000	103.000000	135.000000
bun	4753.0	32.349463	26.792288	1.000000	14.000000	23.000000
urine	4243.0	2191.546047	1455.245777	0.000000	1165.500000	1968.000000
adlp	3464.0	1.157910	1.739672	0.000000	0.000000	0.000000
adls	6238.0	1.637384	2.231358	0.000000	0.000000	1.000000
adlsc	9105.0	1.888272	2.003763	0.000000	0.000000	1.000000

Section 2: Column Renaming

Rename some columns to follow PEP-8 standard

```
Column renames applied:
'd.time' -> 'd_time'
'num.co' -> 'num_co'
```

Total columns: 48

Section 3: Target Variable

Did the patient die within 180 days of study?

To answer this we look at two fields:

- **death** : whether the patient ever died during 180 days (1 = yes, 0 = no)
- **d_time** : days of follow-up (how long they were observed). If this number is greater than 180 days of the trial, that means they survived the trial

death_180d (1 = yes, 0 = no) = {death=1 AND d_time <= 180}

Notice the AND conditional because the patient must've died within the 180 days to satisfy our target. We then use **death_180d** for binary classification

	Outcome	Frequency	Relative Frequency	Percentage (%)
Survived >= 180 days (0)		4840	0.531576	53.157606
Died within 180 days (1)		4265	0.468424	46.842394

Class imbalance ratio (survived: died) = 1.13 : 1

Section 4: Data Type Casting (into Pandas Type)

Correct data types are essential for proper computing and modeling. We apply three categories of correction. Casting to Int8 also handles NaN gracefully:

1. **Binary integer columns** -> Int8 (nullable integer; memory-efficient)
2. **Nominal categorical columns** -> unordered pd.Categorical
3. **Ordinal categorical columns** -> ordered pd.Categorical for proper sorting and comparison

income ordering (ascending socioeconomic status): under \$11k < \$11-\$25k < \$25-\$50k < >\$50k

sfdm2 ordering (best functional status → worst): no disability < moderate ADL impairment < severe SIP impairment < coma/intubated < died before 2-month interview

Data type corrections applied:

death	Int8
hospdead	Int8
diabetes	Int8
dementia	Int8
sex	category
dzgroup	category
dzclass	category
race	category
ca	category
dnr	category
income	category
sfdm2	category

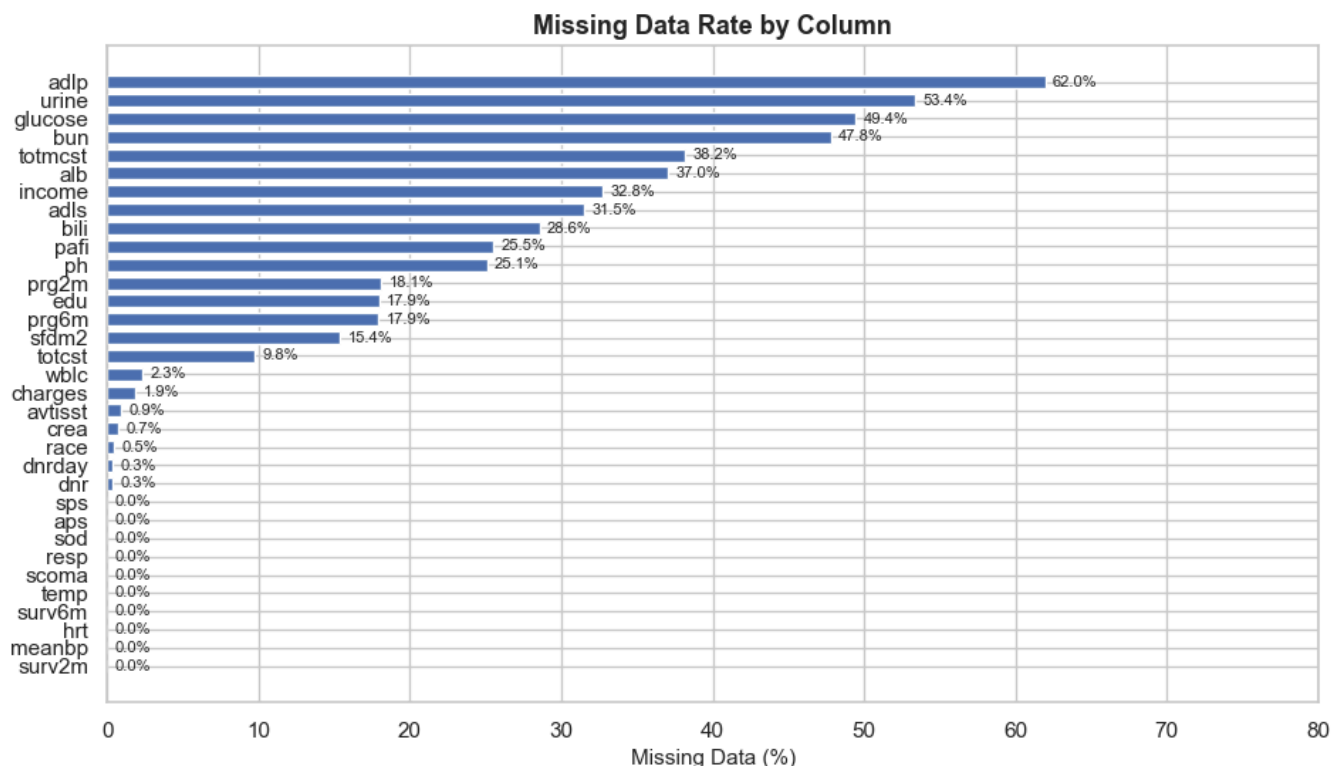
Section 5: Missing Data Analysis

The authors have stated there are a maximum of 5641 N/As. This, along with every other missing data, we must verify and address.

Columns with at least one missing value : 33
Total missing cells : 47,110 / 446,145 (10.6%)

	Missing Count	Missing %
adlp	5641	61.954970
urine	4862	53.399231
glucose	4500	49.423394
bun	4352	47.797913
totmcst	3475	38.165843
alb	3372	37.034596
income	2982	32.751236
adls	2867	31.488193
bili	2601	28.566722
pafi	2325	25.535420
ph	2284	25.085118
prg2m	1649	18.110928
edu	1634	17.946183
prg6m	1633	17.935200
sfdm2	1400	15.376167
totcst	888	9.752883
wblc	212	2.328391
charges	172	1.889072
avtisst	82	0.900604
crea	67	0.735859
race	42	0.461285
dnrday	30	0.329489
dnr	30	0.329489
sps	1	0.010983
aps	1	0.010983
sod	1	0.010983
resp	1	0.010983
scoma	1	0.010983
temp	1	0.010983
surv6m	1	0.010983
hrt	1	0.010983
meanbp	1	0.010983
surv2m	1	0.010983

Max NAs: 5641 (column: 'adlp')



5.1 Missing Data Decision Table

Here we decide how to tackle missing data. Either drop the column entirely or impute mathematically

For most of our biological features that have missing data, we compute the median. The reason is per Agresti et al (2022), chapter 3, that for right-skewed data, using median instead of mean worse the skewness of the data (push it toward the tail). Looking at section 1, our columns the mean > median, which signals right skewed.

Column(s) Decision Rationale

| `adlp` , `adls` | **Drop** | Superseded by `adlsc` per original authors | `dnrday` | **Drop** | Do-no-resus day. Useless for our purpose | `alb` , `pafi` , `bili` , `crea` , `bun` , `wbhc` , `urine` | **imputed values avail** | Per Author's note we have the recommended imputed values | `ph` , `glucose` , `scoma` , `avtisst` , `edu` , `prg2m` , `prg6m` | **Sample median** | high missing | `surv2m` , `surv6m` , `aps` , `sps` | **Sample median** | kept as benchmark columns | `meanbp` , `hrt` , `resp` , `temp` , `sod` | **Sample median** | few missing | `charges` , `totcst` | **Sample median** | Retained for future cost analysis | `totmcst` | **Sample median** | Retained for future cost analysis; many missing values and will be difficult to impute. Contains negative values (balances) | `race` , `income` , `sfdm2` , `dnr` | **Mode imputation** | Most frequent category as best single-value estimate for categoricals |

Benchmark Columns (kept but will be excluded from death_180d model features)

Column	Role	Notes
surv2m , surv6m	SUPPORT model predictions	will be compared against our model
aps	APACHE III severity score	Independent benchmark system widely used in critical care
sps	SUPPORT physiology sub-score	Component of the SUPPORT model ??
prg2m , prg6m	Physician survival estimates	ML vs. clinician judgment comparison
dnr	DNR order status	Should not use as predictor feature as it's reflects human behavior rather than biological factors

Section 6: Feature Pruning

We remove features not used in our analysis:

```
Dropped : 3 columns
Remaining: 9,105 rows x 46 columns
Remaining columns (46):
['id', 'age', 'death', 'sex', 'hospdead', 'slos', 'd_time', 'dzgroup', 'dzclass', 'num_co', 'edu', 'income', 'scoma', 'charges', 'totcst', 'totmcst', 'avtisst', 'race', 'sps', 'aps', 'surv2m', 'surv6m', 'hday', 'diabetes', 'dementia', 'ca', 'prg2m', 'prg6m', 'dnr', 'meanbp', 'wblc', 'hrt', 'resp', 'temp', 'pafi', 'alb', 'bili', 'crea', 'sod', 'ph', 'glucose', 'bun', 'urine', 'sfdm2', 'adlsc', 'death_180d']
```

Section 7: Missing Data Imputation

The original SUPPORT study authors (Harrell, 2022) provide clinically validated **normal fill-in values** for seven physiological variables (see docs/support2_dataset_description.md).

We will use this for imputation

Variable	Normal Fill	Meaning
alb	3.5	Serum albumin - liver / nutrition
pafi	333.3	PaO2/FiO2 ratio - oxygen level
bili	1.01	Bilirubin - liver function
crea	1.01	Creatinine - renal function
bun	6.51	BUN - normal renal function
wblc	9.0	WBC count - immune status
urine	2502	Urine output - renal output

Imputation Summary:

Column	Strategy	Fill Value	Cells Filled
alb	Domain Fill	3.5	3372
pafi	Domain Fill	333.3	2325
bili	Domain Fill	1.01	2601
crea	Domain Fill	1.01	67
bun	Domain Fill	6.51	4352
wblc	Domain Fill	9.0	212
urine	Domain Fill	2502.0	4862
meanbp	Sample Median	77.0	1
hrt	Sample Median	100.0	1
resp	Sample Median	24.0	1
temp	Sample Median	36.6953	1
sod	Sample Median	137.0	1
ph	Sample Median	7.4199	2284
glucose	Sample Median	135.0	4500
scoma	Sample Median	0.0	1
avtisst	Sample Median	19.5	82
edu	Sample Median	12.0	1634
surv2m	Sample Median	0.7159	1
surv6m	Sample Median	0.575	1
aps	Sample Median	34.0	1
sps	Sample Median	23.8984	1
prg2m	Sample Median	0.7	1649
prg6m	Sample Median	0.5	1633
charges	Sample Median	25024.0	172
totcst	Sample Median	14452.7344	888
totmcst	Sample Median	13223.5	3475
race	Sample Mode	white	42
income	Sample Mode	under \$11k	2982
sfdm2	Sample Mode	<2 mo. follow-up	1400
dnr	Sample Mode	no dnr	30

Total cells imputed: 38,572

Imputation complete, zero missing values remain in any column.

Dataset shape after imputation: 9,105 rows x 46 columns

Section 8: Defining Feature, Outcome Separation, Leakage

Category	Columns
Target	death_180d
Outcome columns (excluded due to result leakage)	death , hospdead , d_time , slos
Row identifier	id
Features	All remaining columns

Section 9: Post-Cleaning Summary

We compare the dataset before and after cleaning and present the five-number summary for key physiological lab variables to confirm that imputation and outlier retention produced a coherent, analysis-ready dataset.

```
=== Pre-Cleaning ===
Shape      : 9,105 rows x 48 columns
Missing cells : 47,110

=== Post-Cleaning ===
Shape      : 9,105 rows x 46 columns
Missing cells : 0
Rows retained : 100.0%
```

9.1 Important Lab Variables Summary (Post-Cleaning)

	Min	Q1	Median	Q3	Max	Mean	
age	18.041990	52.797000	64.856990	73.998960	101.84796	62.650823	1
meanbp	0.000000	63.000000	77.000000	107.000000	195.00000	84.545579	2
hrt	0.000000	72.000000	100.000000	120.000000	300.00000	97.157023	3
temp	31.699220	36.195310	36.695310	38.195310	41.69531	37.103296	
alb	0.399963	2.699707	3.500000	3.500000	29.00000	3.154017	
bili	0.099991	0.599976	1.010000	1.299805	63.00000	2.113261	
crea	0.099991	0.899902	1.199951	1.899902	21.50000	1.765361	
glucose	0.000000	133.000000	135.000000	136.000000	1092.00000	147.580121	6
bun	1.000000	6.510000	6.510000	24.000000	300.00000	19.998739	2
urine	0.000000	2075.000000	2502.000000	2502.000000	9000.00000	2357.326071	100

Section 10: Save Cleaned Dataset

Cleaned dataset saved to : C:\Users\mwee\Documents\USD-AAI\aa-500\final\usd-aa-500-final\data\support2_cleaned.csv
Final shape : 9,105 rows x 46 columns

Column list and dtypes:

id	int64
age	float64
death	Int8
sex	category
hospdead	Int8
slos	int64
d_time	int64
dzgroup	category
dzclass	category
num_co	int64
edu	float64
income	category
scoma	float64
charges	float64
totcst	float64
totmcst	float64
avtisst	float64
race	category
sps	float64
aps	float64
surv2m	float64
surv6m	float64
hday	int64
diabetes	Int8
dementia	Int8
ca	category
prg2m	float64
prg6m	float64
dnr	category
meanbp	float64
wblc	float64
hrt	float64
resp	float64
temp	float64
pafi	float64
alb	float64
bili	float64
crea	float64
sod	float64
ph	float64
glucose	float64
bun	float64
urine	float64
sfdm2	category
adlsc	float64
death_180d	int64

SUPPORT2 Dataset: Exploratory Data Analysis

AAI-500 Applied Statistics for AI | Final Team Project

Notebook Objectives

This notebook covers the **Exploratory Data Analysis** phase of the project pipeline:

1. Detect Outliers
 2. Univariate Analysis
 3. Bivariate Analysis
 4. Demographic Analysis
 5. Disease Group Analysis
 6. DNR Status Analysis
-

Learnings from the 1990s study

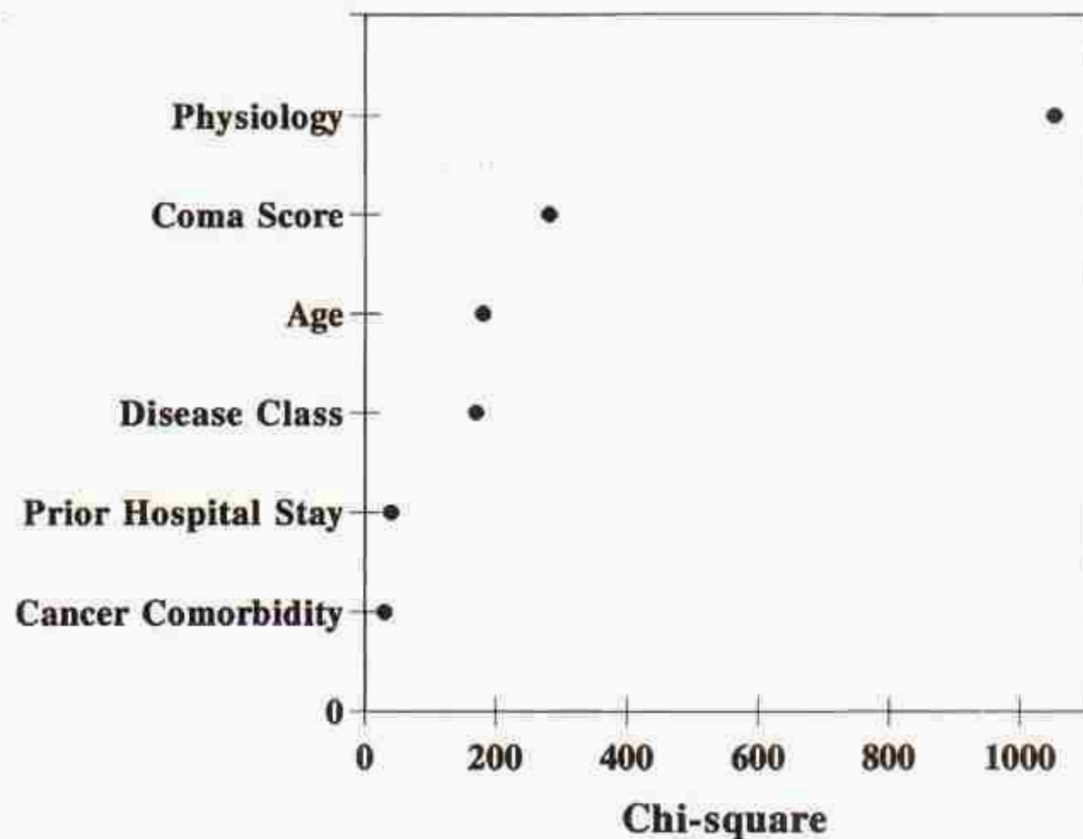


Figure 1. Relative contributions of major prognostic elements in the SUPPORT model as measured by the amount of chi-square accounted for by each element. Physiology = all physiologic measures except Glasgow coma scale; cancer comorbidity = presence of cancer in addition to disease category; previous hospital stay = days in the hospital before study entry.

To inform our work, we review Knaus et al's (1995) paper. According to Knaus et al these features were most weighted in their model:

- Physiology Score (calculated from wbc, albumin, etc). Some salient ones are Blood Pressure and Oxygenation levels
- Neurological Function (Glasgow coma scale)
- Patient age (but only when coupled with certain diseases like COPD)
- Apache III score
- Disease-Specific Interactions: The mathematical weight of certain features was explicitly programmed to vary based on the patient's primary disease: Albumin x Disease: A low albumin level strongly predicted death in cancer, COPD, and congestive heart failure, but was relatively unimportant in coma or MOSF. WBC x Disease: A low WBC count was specifically associated with a greater risk of death in acute respiratory failure and MOSF. Age x Disease: Advancing age heavily impacted short-term survival in

COPD but had almost no incremental effect on patients with severe acute complications like MOSF or malignancy.

Section 0: Setup & Imports

Section 1: Load Cleaned Data

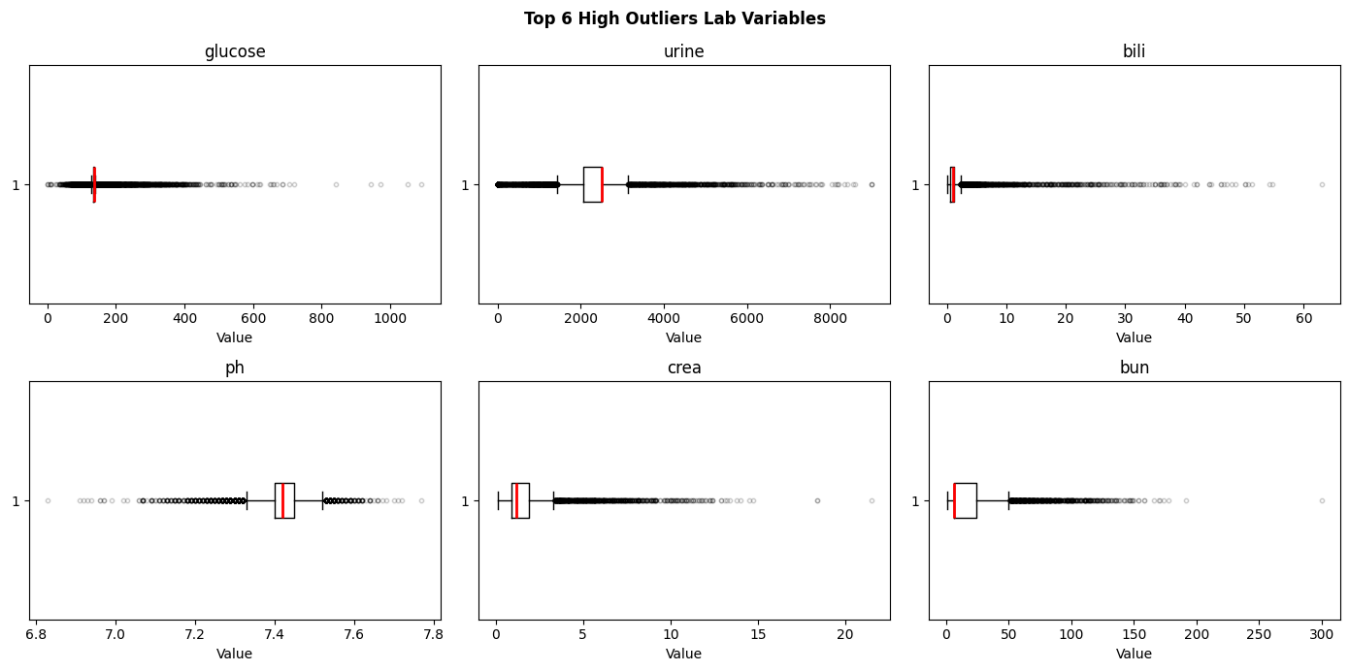
```
Shape: 9,105 rows x 46 columns
Survived 180d : 4,840 (53.2%)
Died within 180d: 4,265 (46.8%)
```

Section 2: Physiology Outlier Detection (IQR Method)

```
IQR = Q3 - Q1
Lo whisker = Q1 - 1.5IQR
Hi whisker = Q3 + 1.5IQR
```

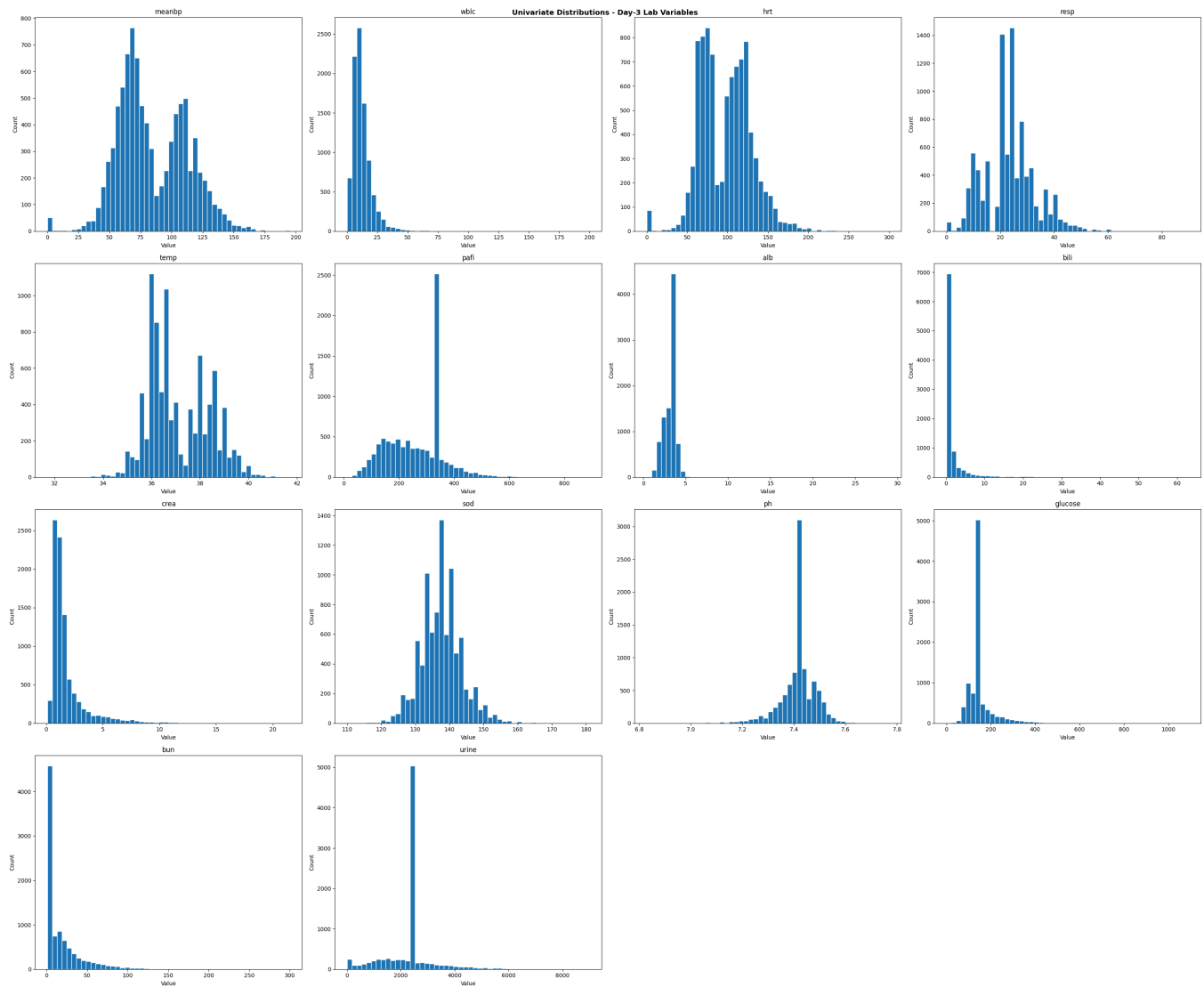
- Urine boxplot median is pushed all the way right because of imputation with Norm-Fill.
- Glucose boxplot collapses due to high number of imputed NAs.
- **Probably better to run outliers before imputation**

Column	Lo Whisker	Hi Whisker	Outliers	Outlier %
glucose	128.500	140.500	4323	47.5
urine	1434.500	3142.500	2361	25.9
bili	-0.450	2.350	1397	15.3
ph	7.325	7.524	1107	12.2
crea	-0.600	3.400	987	10.8
bun	-19.725	50.235	916	10.1
wblc	-5.149	27.249	431	4.7
resp	3.000	43.000	313	3.4
sod	123.500	151.500	256	2.8
alb	1.499	4.700	154	1.7
pafi	-49.950	563.250	59	0.6
hrt	0.000	192.000	40	0.4
temp	33.195	41.195	14	0.2
meanbp	-3.000	173.000	6	0.1

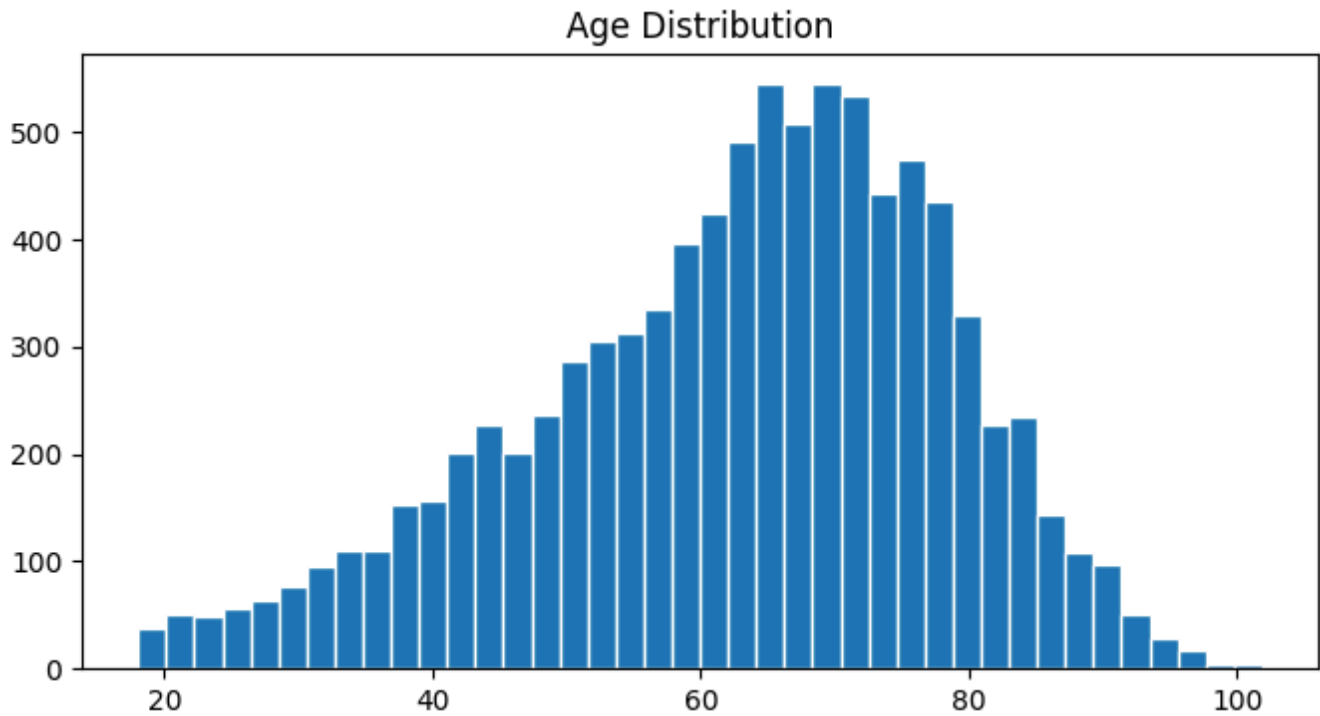


Section 3: Univariate Distributions

- Lab features are mainly right skewed
- Some imputations cause spikeness (glucose, pafi, bili)

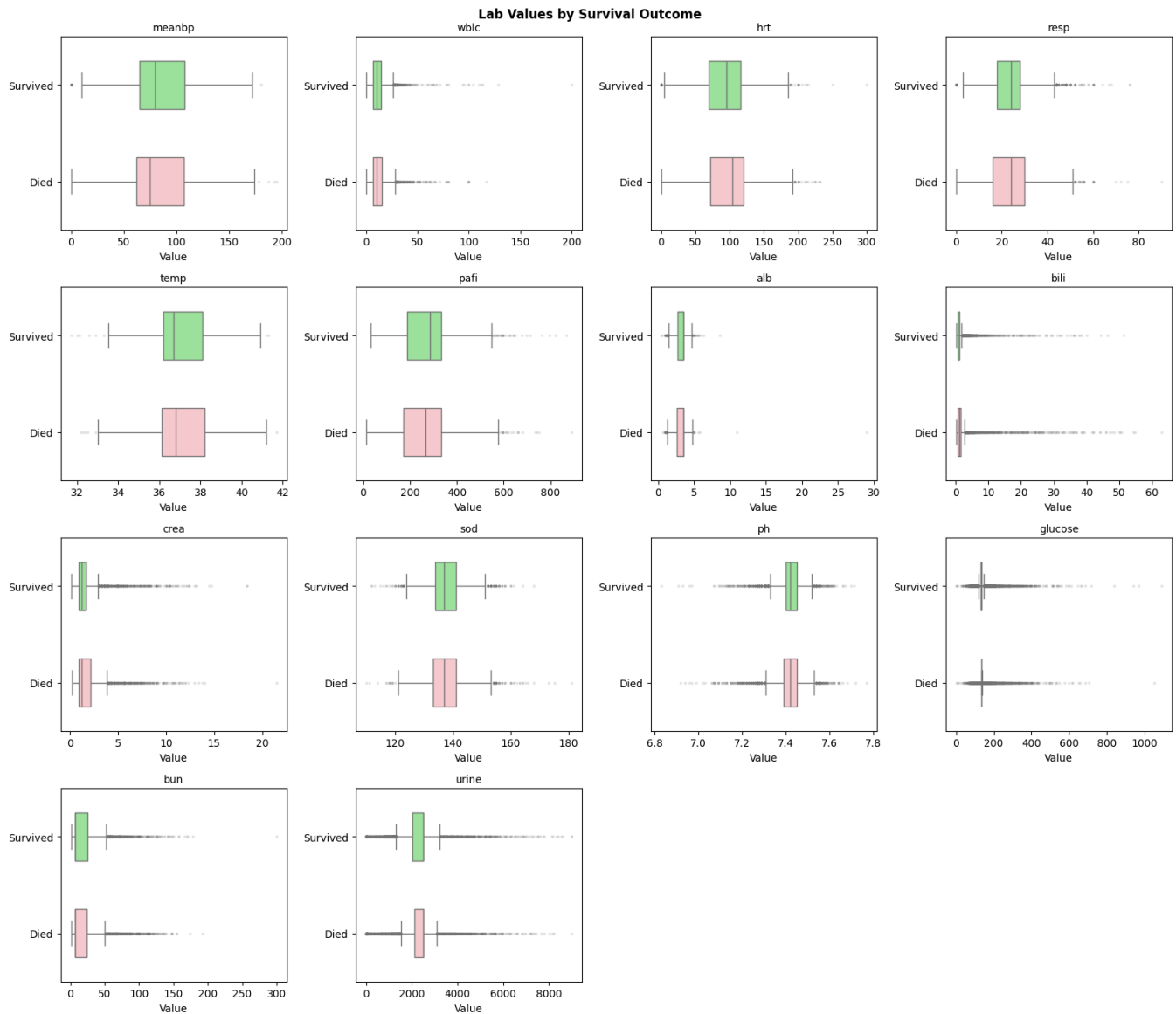


3.1 Age follows normal distribution



Section 4: Bivariate Analysis - Lab Values by Survival Outcome

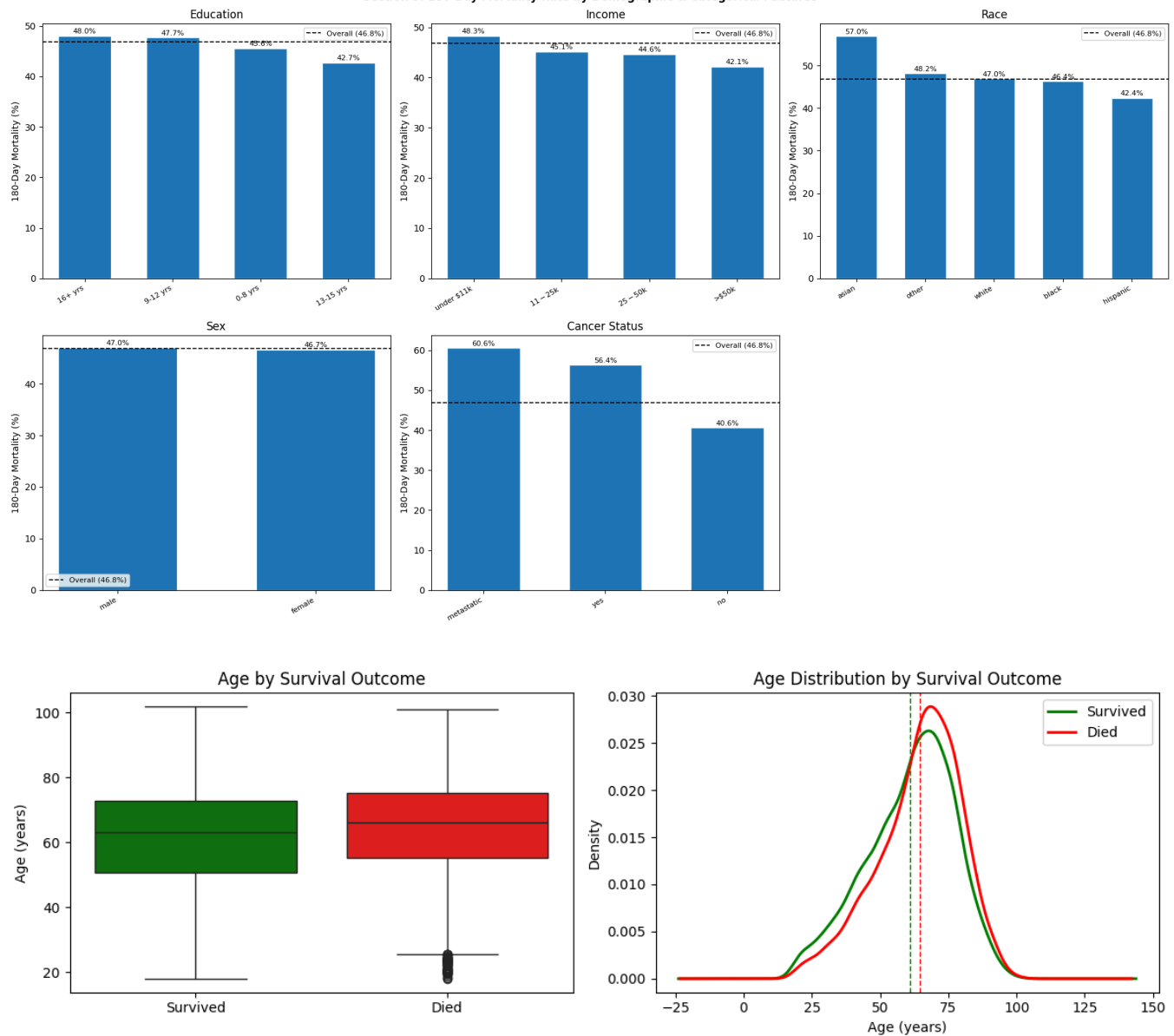
- Difficult to see any impact these lab values make with this analysis
- The paper did say this only becomes evident when combined with Disease group. For example, a high creatinine with kidney failure gives high likelihood of mortality.



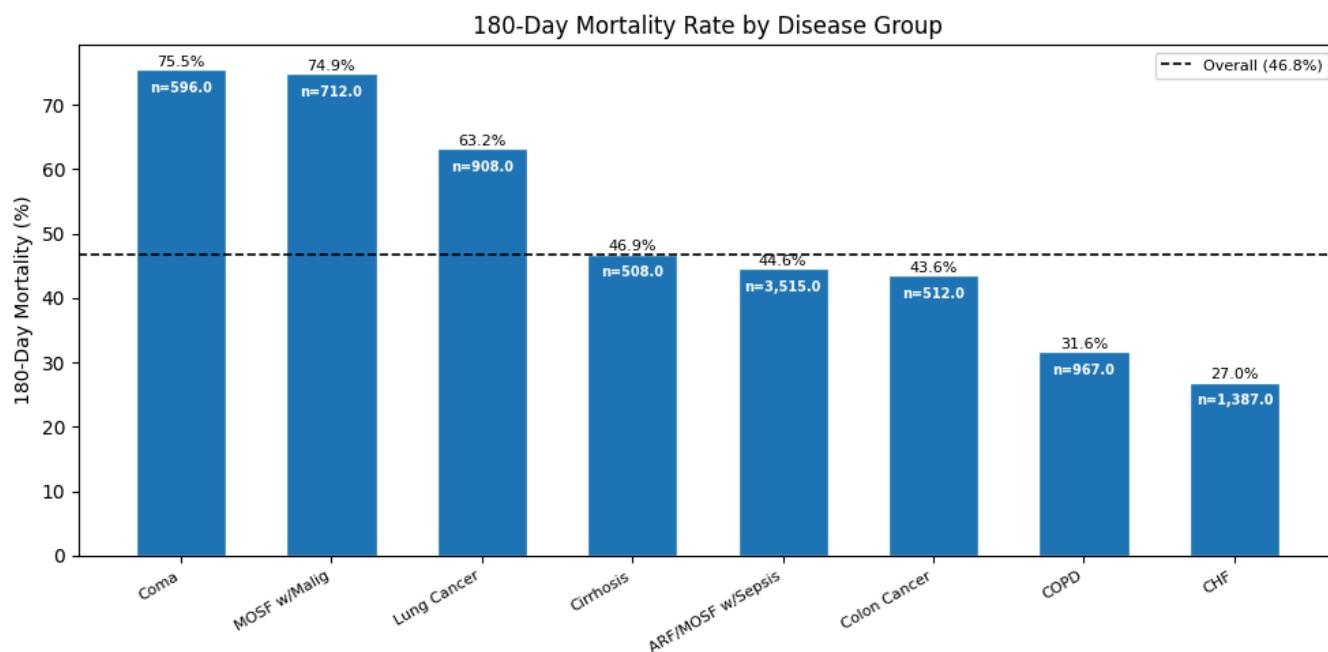
Section 5: Demographic Analysis

- Mortality gradual decline as wealth increases
- Asian group a bit more mortality

Section 5: 180-Day Mortality Rate by Demographic & Categorical Features

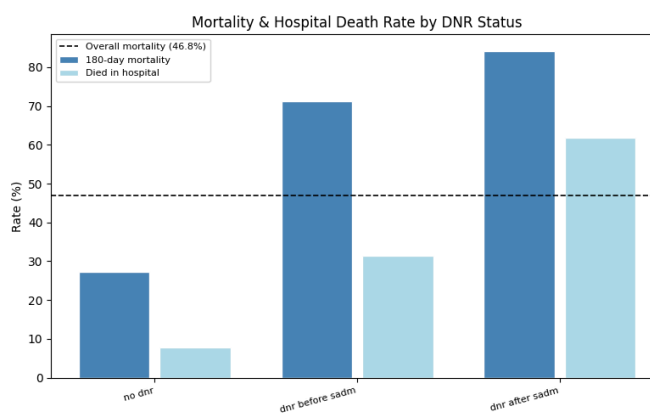
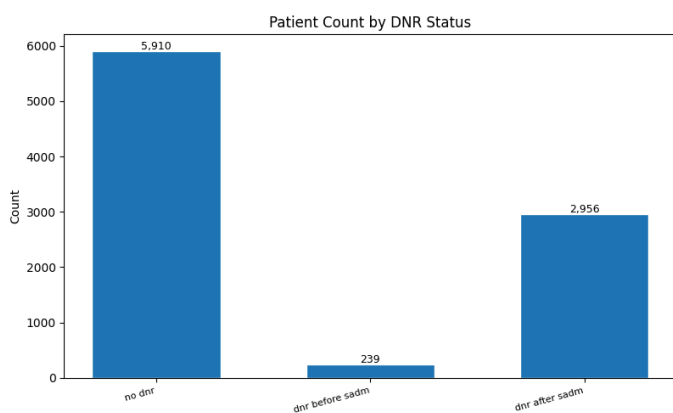


Section 6: Disease Group Analysis



Section 7: DNR Status

- **no dnr**: no order placed
- **dnr before sadm**: order existed before study admission
- **dnr after sadm**: order placed during the admission (possibly due to deteriorating conditions)



DNR Order Rate by Disease Group

